

STAT 516: Homework 7 Answer Key

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2}e^{-y}, & -y < x < y, \ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P(X < 1 \mid Y = 3).$$

Solution.

First, find the marginal PDF of Y :

$$f_Y(y) = \int_{-y}^y \frac{1}{2}e^{-y} dx = \frac{1}{2}e^{-y}(2y) = ye^{-y}.$$

Thus, the conditional PDF of X given $Y = y$ is

$$f_{X|Y=y}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{\frac{1}{2}e^{-y}}{ye^{-y}} = \frac{1}{2y}, \ -y < x < y.$$

Given $Y = 3$,

$$f_{X|Y=3}(x) = \frac{1}{6}, \quad -3 < x < 3.$$

Therefore,

$$P(X < 1 \mid Y = 3) = \int_{-3}^1 \frac{1}{6} dx.$$

Hence,

$$P(X < 1 \mid Y = 3) = \frac{1 - (-3)}{6} = \frac{4}{6} = \frac{2}{3}.$$

2. Let X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2x, & 0 < x < 1, \ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P\left(X + Y \leq 1 \mid X \leq \frac{1}{2}\right).$$

Solution.

By the definition of conditional probability,

$$P\left(X + Y \leq 1 \mid X \leq \frac{1}{2}\right) = \frac{P\left(X + Y \leq 1, X \leq \frac{1}{2}\right)}{P\left(X \leq \frac{1}{2}\right)}.$$

First,

$$P\left(X \leq \frac{1}{2}\right) = \int_0^{1/2} \int_0^1 2x \, dy \, dx = \int_0^{1/2} 2x \, dx = [x^2]_0^{1/2} = \frac{1}{4}.$$

For the numerator, the conditions are

$$0 < x \leq \frac{1}{2} \quad \text{and} \quad 0 < y \leq 1 - x.$$

Therefore,

$$P\left(X + Y \leq 1, X \leq \frac{1}{2}\right) = \int_0^{1/2} \int_0^{1-x} 2x \, dy \, dx.$$

Thus,

$$P\left(X + Y \leq 1, X \leq \frac{1}{2}\right) = \int_0^{1/2} 2x(1-x) \, dx = \int_0^{1/2} (2x - 2x^2) \, dx.$$

Hence,

$$P\left(X + Y \leq 1, X \leq \frac{1}{2}\right) = \left[x^2 - \frac{2}{3}x^3\right]_0^{1/2} = \frac{1}{4} - \frac{1}{12} = \frac{1}{6}.$$

Finally,

$$P\left(X + Y \leq 1 \mid X \leq \frac{1}{2}\right) = \frac{1/6}{1/4} = \frac{2}{3}.$$

3. Let X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 24xy, & 0 < x < 1, \, 0 < y < 1-x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P\left(Y < X \mid X = \frac{1}{3}\right).$$

Solution.

First, find the marginal PDF of X :

$$f_X(x) = \int_0^{1-x} 24xy \, dy, \quad 0 < x < 1.$$

Therefore,

$$f_X(x) = 24x \left[\frac{y^2}{2} \right]_0^{1-x} = 12x(1-x)^2.$$

At $x = \frac{1}{3}$,

$$f_X\left(\frac{1}{3}\right) = 12\left(\frac{1}{3}\right)\left(1 - \frac{1}{3}\right)^2 = \frac{16}{9}.$$

The conditional PDF of Y given $X = \frac{1}{3}$ is

$$f_{Y|X=1/3}(y) = \frac{f_{X,Y}\left(\frac{1}{3}, y\right)}{f_X\left(\frac{1}{3}\right)}.$$

Therefore,

$$f_{Y|X=1/3}(y) = \frac{24\left(\frac{1}{3}\right)y}{16/9} = \frac{8y}{16/9} = \frac{9}{2}y,$$

for

$$0 < y < 1 - \frac{1}{3} = \frac{2}{3}.$$

We want

$$P\left(Y < X \mid X = \frac{1}{3}\right) = P\left(Y < \frac{1}{3} \mid X = \frac{1}{3}\right).$$

Thus,

$$P\left(Y < X \mid X = \frac{1}{3}\right) = \int_0^{1/3} \frac{9}{2}y \, dy = \frac{9}{2} \left[\frac{y^2}{2} \right]_0^{1/3}.$$

Hence,

$$P\left(Y < X \mid X = \frac{1}{3}\right) = \frac{9}{4} \left(\frac{1}{9} \right) = \frac{1}{4}.$$

4. A fair coin is tossed. If a head occurs, one fair six-sided die is rolled. If a tail occurs, two independent fair six-sided dice are rolled.

Let Y be the total number of points showing on the die or dice.

Calculate

$$E[Y].$$

Solution.

Let H denote the event that the coin lands heads, and let T denote the event that the coin lands tails.

Since the coin is fair,

$$P(H) = P(T) = \frac{1}{2}.$$

If a head occurs, one die is rolled. Therefore,

$$E[Y | H] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{7}{2}.$$

If a tail occurs, two independent dice are rolled. Therefore,

$$E[Y | T] = 2 \left(\frac{7}{2} \right) = 7.$$

By the law of total expectation,

$$E[Y] = E[Y | H] P(H) + E[Y | T] P(T).$$

Thus,

$$E[Y] = \frac{7}{2} \left(\frac{1}{2} \right) + 7 \left(\frac{1}{2} \right) = \frac{21}{4} = 5.25$$

5. Let X and Y denote the values of two stocks at the end of a five-year period.

Suppose

$$X \sim \text{Uniform}(0, 12).$$

Given $X = x$, suppose

$$Y | X = x \sim \text{Uniform}(0, x).$$

Calculate

$$\text{Cov}(X, Y).$$

Solution.

Since

$$Y | X = x \sim \text{Uniform}(0, x),$$

we have

$$E[Y | X = x] = \frac{0 + x}{2} = \frac{x}{2}.$$

Therefore,

$$E[Y | X] = \frac{X}{2}.$$

Using the law of total expectation,

$$E[Y] = E[E[Y | X]] = E\left[\frac{X}{2}\right] = \frac{1}{2} E[X].$$

Since

$$X \sim \text{Uniform}(0, 12),$$

we have

$$E[X] = \frac{0 + 12}{2} = 6.$$

Thus,

$$E[Y] = 3.$$

Next,

$$E[XY] = E[E[XY | X]].$$

Given X , the variable X is fixed, so

$$E[XY | X] = X E[Y | X].$$

Therefore,

$$E[XY] = E\left[X \cdot \frac{X}{2}\right] = \frac{1}{2} E[X^2].$$

For

$$X \sim \text{Uniform}(0, 12),$$

$$E[X^2] = \frac{12^2}{3} = 48.$$

Thus,

$$E[XY] = \frac{1}{2}(48) = 24.$$

Finally,

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y].$$

Therefore,

$$\text{Cov}(X, Y) = 24 - (6)(3) = 6.$$

6. Let X be the loss amount for a policyholder. Suppose

$$X \sim \text{Uniform}(0, b).$$

The policy has a deductible of \$180, which means the policyholder is responsible for paying the first \$180 of each loss.

Therefore, the unreimbursed portion of a loss is

$$U = \min(X, 180).$$

If

$$E[U] = 144,$$

calculate b .

Solution.

Since the deductible is \$180,

$$U = \min(X, 180).$$

Assuming $b > 180$,

$$E[U] = \int_0^{180} x f_X(x) dx + \int_{180}^b 180 f_X(x) dx.$$

Since

$$X \sim \text{Uniform}(0, b),$$

$$f_X(x) = \frac{1}{b}, \quad 0 \leq x \leq b.$$

Therefore,

$$E[U] = \frac{1}{b} \int_0^{180} x dx + \frac{1}{b} \int_{180}^b 180 dx.$$

Thus,

$$E[U] = \frac{1}{b} \left[\frac{x^2}{2} \right]_0^{180} + \frac{180}{b} [x]_{180}^b = \frac{16200}{b} + 180 - \frac{32400}{b} = 180 - \frac{16200}{b}.$$

Given

$$E[U] = 144.$$

Therefore,

$$144 = 180 - \frac{16200}{b}.$$

Solving for b ,

$$b = \frac{16200}{36} = 450.$$

7. Let

$$E[Y | X = x] = 2x, \quad \text{Var}(Y | X = x) = 4x^2,$$

and suppose

$$X \sim \text{Uniform}(0, 1).$$

Calculate

$$\text{Var}(Y).$$

Solution.

$$\begin{aligned}\text{Var}(Y) &= \text{Var}(\text{E}[Y \mid X]) + \text{E}[\text{Var}(Y \mid X)] \\&= \text{Var}(2X) + \text{E}[4X^2] \\&= 4 \text{Var}(X) + 4 \text{E}[X^2] \\&= 4 \left(\frac{1}{12} \right) + 4 \left(\frac{1}{3} \right) \\&= \frac{1}{3} + \frac{4}{3} \\&= \frac{5}{3}.\end{aligned}$$